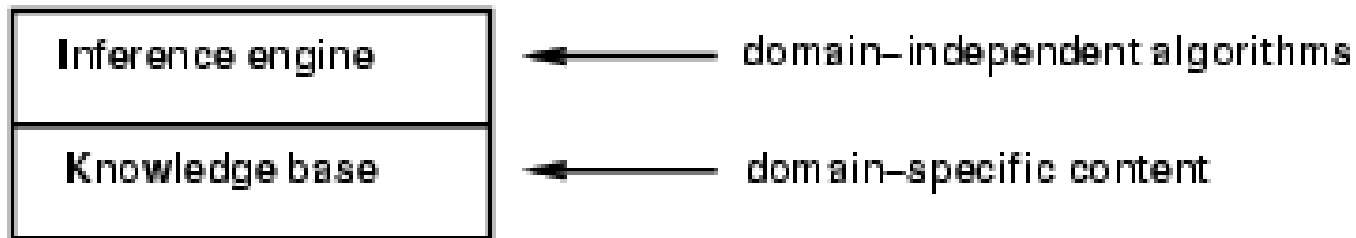


CHAPTER FOUR

KNOWLEDGE REPRESENTATION AND REASONING - I **KNOWLEDGE REPRESENTATION AND INFERENCING USING PREPOSITIONAL LOGIC**

Knowledge Base Agent

- Knowledge base agent is an agent that perform action using the knowledge it has and reason about their action using its inference procedure.



- Knowledge base is a set of representation of facts and their relation ships called **rules** about the world
- Each fact/rules are called **sentences** which is represented using a language called knowledge representation language.
- **Declarative approach to building an agent (or other system):**
 - Tell it what it needs to know (Knowledge base)
 - Ask what it knows
- **Answers should follow from the KB**

Knowledge Bases Agent

- The agent must be able to:
 - ▶ Represent states of the world, actions, etc.
 - ▶ Incorporate new percepts (facts and rules)
 - ▶ Deduce hidden properties of the world
 - ▶ Deduce appropriate actions
 - ▶ Update internal representations of the world

Knowledge Bases Agent

- Representation of state of the world using KB
 - Example of KB written in that represent a state of a world in which only Azieb is living (PROLOG representation)
 - Moreover, the required information are gender and age given name
 - FACTS
 1. female(azieb).
 2. age(azieb, 18).

Example

- Agent should incorporate new percept
 - For example,
 1. Azieb get married with melaku, they born two kids selam and solomon
 2. This can be reflected by incorporating the new percepts
 - FACTS
 1. female(azieb).
 2. Age(azieb, 18).
 3. Married(azieb, melaku).
 4. male(melaku).
 5. female(selam).
 6. male(solomon).
 7. parent(melaku,selam).
 8. parent(azieb,selam).
 9. parent(melaku,solomon).
 10. parent(azieb,solomon).
 11. age(melaku,26).
 12. age(selam, 5).
 13. age(solomon, 1).
 - RULE
 1. married(X,Y):-married(Y,X).
 2. father(X,Y):-male(X),parent(X,Y).
 3. mother(X,Y):-female(X),parent(X,Y).
 4. wife(X,Y):-parent(X,Z),parent(Y,Z).
 5. brother(X,Y):-male(X), parent(Z,X),parent(Z,Y).
 6. sister(X,Y):-female(X), parent(Z,X),parent(Z,Y).

Example

- The agent should deduce hidden portion of the world like azieb and selam are beautiful.
 - If one asks questions like
 - ?brother(aster,belay).
 - Married(melaku, azieb).
 - The system must respond correctly
- Deduce appropriate action to query.
 - If one asks questions like
 - who is the sister of whom,
 - who is the father of belay,
 - who is the mother of Selam, etc
 - The system must respond correctly
- Agent should update internal representation of the world.
 - For example, if died(melaku) is given we should modify any fact that tells us about live history about melaku.

Knowledge representation

- Knowledge representation refers to the technique how to express the available facts and rules inside a computer so that agent will use it to perform well.
- Knowledge representation consists of:
 - ▶ **Syntax (grammar)**: possible physical configuration that constitute a sentence (fact or rule) inside the agent architecture.
 - ▶ For example one possible syntax rule may be every sentence must end with full stop.
 - ▶ **Semantics (concept)**: determine the facts in the world to which the sentence refers
 - ▶ Without semantics a sentence is just a sequence of characters or binary sequences
 - ▶ Semantic defines the meaning of the sentence

Knowledge representation

- E.g., In the language of arithmetic
 - ▶ $x+2 \geq y$ is a sentence; $x^2+y > \{\}$ is not a sentence
 - ▶ $x+2 \geq y$ is true iff the number $x+2$ is not less than the number y
 - ▶ $x+2 \geq y$ is true in a world where $x = 7, y = 1$
 - ▶ $x+2 \geq y$ is false in a world where $x = 0, y = 6$
- Given clear definition of semantics and syntax of a language, we call that language **logical language**.
- Knowledge representation is used to represent part of the world (facts and their association) into ideal computer system
- KB for agent program can be represented using programming language designed for this purpose like LISP and PROLOG

Logic

- **Logic** is the study of the principles of reasoning and arguments towards the truth of a given conclusion given premises.
- **Logic** in AI is the key idea for KB design, KB representation and inferencing (reasoning)
- **Logic is** formal languages use for representing information so that conclusions can be drawn
- Logic is the systematic study of the general conditions of valid inferences

Types of Logic

- In mathematics there are different kinds of logics.
- Some of these according to order of their generality are
 - Propositional logic
 - First order logic
 - Second order logic and more
- First order logic can be used to design, represent or infer for any environment in the real world.

Propositional (Boolean) logic (PL)

- Proposition is statement which is either true or false but not both at any time.
- A statement is a sentence which is either true or false.
- PL uses declarative sentences only
- PL doesn't involve quantifiers.
- Not all sentences are statement (interrogatives, imperatives and exclamatory)
- Proposition can be conditional or unconditional
- **Examples**
 - Socrates is mortal (unconditional)
 - If the winter is severe, students will not succeed. (conditional)
 - All are the same iff their color is black (conditional)
- In propositional logic, symbols represent the whole proposition.
- **Examples:**
 - M = Socrates is mortal
 - W = winter is severe
 - S = students will not succeed

Propositional (Boolean) logic (PL)

- Proposition symbols can be combined using **Boolean connectives** to generate new proposition with complex meaning
- Symbols involved in PL:
 - Logical constants (**TRUE** and **FALSE**)
 - Proposition symbols (also called atomic symbols) like M, W, S
 - Logical connectives
 - $\neg$ (**negation**),
 - $\wedge$ (**conjunction**),
 - $\vee$ (**disjunction**),
 - $\Leftrightarrow$ (**bi-implication or equivalence**),
 - $\Rightarrow$ (**implication**) and **parenthesis**

Propositional (Boolean) logic (PL)

➤ Rules

- Logical constants and propositional symbols are sentence by them selves
- Wrapping parenthesis around a sentence yield a sentence like $(P \vee Q)$
- **Literal** are atomic symbols or negated atomic symbols
- **Complex sentence** can be formed by combining simpler sentences with logical connectors

PL connector priority

- Priority of logical connectives from **highest** to **lowest**
 - ▶ parenthesis
 - ▶ Negation
 - ▶ Conjunction
 - ▶ Disjunction
 - ▶ Implication
 - ▶ Bi-implication

Types of sentence

- Given a sentence α , this sentence according to the world considered can be
 - Valid (tautology)
 - Invalid (contradiction)
 - Satisfiable (neither valid nor invalid)

Validity (tautology)

- A sentence is valid iff it is true under any interpretations in all possible world
 - ▶ Example: $x > 4$ or $x \leq 4$;
 - ▶ Water boils at 100 degree centigrade
 - ▶ Human has two legs (may not be valid)
 - ▶ Books have page number (may not be valid)

Satisfiability

- A sentence is **satisfiable** iff there is some interpretation in some world for which it is true.
- Every valid sentence is satisfiable
 - Example: $x+2 = 20$
 - Every student of AI are in their class
- A sentence which is not satisfiable is unsatisfiable (contradiction or invalid).

Entailment

- Entailment means that one thing follows from another:
- It can be represented by $\models$ symbol (double turnstyle)
- $KB \models \alpha$ shows α can be entailed from KB
- Knowledge base KB entails sentence α if and only if α is true in all worlds where KB is true
 - E.g., the KB containing “the Giants won” and “the Reds won” entails “Either the Giants won or the Reds won”
 - E.g., $x+y = 4$ entails $4 = x+y$
 - E.g., $x+y = 4$ can't entail $x=2$ and $y=2$
 - Entailment is a relationship between sentences (i.e., syntax) that is based on semantics

Inference Procedure

- An inference procedure is a procedure used as reasoning engine.
- It can do:
 1. Given KB, generate new sentence α that can be entailed by KB and we call the inference procedure entail α
 2. Given KB and α , it will prove whether α is entailed by KB or not
- $KB \vdash_i \alpha$ means sentence α can be derived from KB by procedure i ($\vdash$ is called **turnstile** or **single turnstyle**)

Inference Procedure property

- **Soundness**: inference procedure i is said to be sound:
if whenever $KB \vdash_i \alpha$, it is also true that $KB \models \alpha$
- **Completeness**: inference procedure i is said to be complete if
whenever $KB \models \alpha$, it is also true that $KB \vdash_i \alpha$

Inference Procedure property

- The record of operation of a sound inference procedure is called a **proof**
- Soundness of an inference can be established through truth table.
- For example and inference procedure that entails P from a KB which consists of $P \rightarrow Q$ & Q is not sound as shown below

	P	Q	$P \rightarrow Q$	Remark
1	T	T	T	Q, $P \rightarrow Q$, & P are true
2	T	F	F	Premises doesn't satisfied
3	F	T	T	<i>Premises satisfied but not the conclusion</i>
4	F	F	T	Premises doesn't satisfied

Rules of inference for PL

- There are a set of already identified sound inference rules.
- These are:
 1. Modes Ponens or implication elimination
 2. And Elimination
 3. And introduction
 4. Or introduction
 5. Double negation elimination
 6. Unit resolution
 7. Resolution
- We will discuss all this one by one in the next slides after discussing normal forms

Normal Forms of Logical expression

- There are different standard forms of expressing PL statement.
- Some inference procedure require their knowledge base to be in one of the normal forms to infer from it.
- Hence, we need to convert the logical expression which are the source of knowledge into the appropriate forms
- Some of the normal forms are stated bellow

Normal Forms of Logical expression

▶ Clausal normal form:

- ▶ it is a set of one or more literals connected with the disjunction operator (disjunction of literals).
- ▶ It doesn't allow negation marker except as a prefix to a symbol
Example $\sim P \vee Q \vee \sim R$ is a clausal form
- ▶ According to the definition a single symbol or negation of a symbol is in clausal form

▶ Conjunctive normal forms (CNF):

- ▶ Simply defined as a sequences of one or more clauses connected by the conjunction operator
- ▶ Or it is conjunction of clauses
- ▶ Or conjunction of disjunction of literals
Example $(A \vee B) \wedge (C \vee D)$
- ▶ Note: connector is needed if we have two or more elementary unit to connect to each other to form complex expression
- ▶ This is the basis for the generalized resolution inference procedure

Normal Forms of Logical expression

- ▶ Disjunctive normal form (DNF):
 - ▶ disjunction of conjunction of literals.
Example $(A \wedge B) \vee (C \wedge D)$
- ▶ Horn form: conjunction of literals implies a literal.
Example $(A \wedge B \wedge C \wedge D) \Rightarrow E$

Inference procedure and normal forms

- The inference procedure that we have seen before are all sound
- If KB is represented in CNF, the generalized resolution inference procedure is complete
- If KB is represented in Horn form, the generalized modes ponens algorithm is complete
- It can be proved that every sentence of human language can be represented using logic as CNF. However, it is not possible in Horn form.
- Therefore, CNF is a more powerful representation technique for knowledge
- But, Horn form representation of knowledge is easily understandable and convenient. It also require polynomial time inference procedure

Modes Ponens

➤ There are a set of already identified **inference rules**. These are:

1. Modes Ponens or implication elimination

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

2. And Elimination

3. And introduction

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_N}{\alpha_i}$$

4. Or introduction

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_N}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_N}$$

5. Double negation elimination

$$\frac{\alpha_i}{\alpha_1 \vee \alpha_2 \vee \dots \vee \alpha_N}$$

6. Unit resolution

$$\frac{\neg \neg \alpha}{\alpha}$$

7. Resolution

$$\frac{\alpha \vee \beta, \neg \beta}{\alpha}$$

$$\frac{\alpha \vee \beta, \neg \beta \vee \gamma}{\alpha \vee \gamma} \quad \text{or equivalently} \quad \frac{\neg \alpha \Rightarrow \beta, \beta \Rightarrow \gamma}{\neg \alpha \Rightarrow \gamma}$$

Generalized Resolution for PL

- Given any two clauses A and B, if there are any literal P_1 in A which has a complementary literal P_2 in B, delete P_1 and P_2 from A and B and construct a disjunction of the remaining clauses.

- The clause constructed is called the resolvent of A and B.

- ▶ For example, consider the following clauses

$$A: \quad P \vee Q \vee R$$

$$B: \quad \sim P \vee Q \vee M$$

$$C: \quad \sim Q \vee S$$

- From clause A and B, if we remove P and $\sim P$, it resolves into clause like
 $D: Q \vee R \vee Q \vee M \equiv Q \vee R \vee M.$

- If Q of clause D and $\sim Q$ of clause C resolved, we get clause
E: $R \vee M \vee S$

- **Similarly**

$$F: P \vee R \vee S \quad \text{from A and C}$$

$$G: Q \vee M \vee R \vee S \quad \text{from F and B etc}$$

Generalized Resolution for PL

- Resolution can also be used to prove a sentence whether it is valid or not.
- That can be done using direct method or indirect method (refutation)
- For example Given a KB consists of the sentences
 - A: $P \vee Q \vee R$
 - B: $\sim P \vee R$
 - C: $\sim Q$
- We need to prove R
- **Method One (Direct)**
 - **Entail D:** $Q \vee R$ from A and B
 - **Entail E:** R from D and C

Note: in order to apply resolution for proving a theory, make sure first all the knowledge is in its clausal form

Generalized Resolution for PL

➤ Method Two (refutation)

- Insert the negation of the statement as valid sentence

A: $P \vee Q \vee R$

B: $\sim P \vee R$

C: $\sim Q$

D: $\sim R$

- Find fallacy in the KB that will indicate the added statement is invalid so that its negation is valid

Entail E: $Q \vee R$ from A and B

Entail F: R from C and E

Entail G: $\{\}$ from D and F

An empty clause shows **false**. This proves the contradiction.

Generalized Resolution for PL

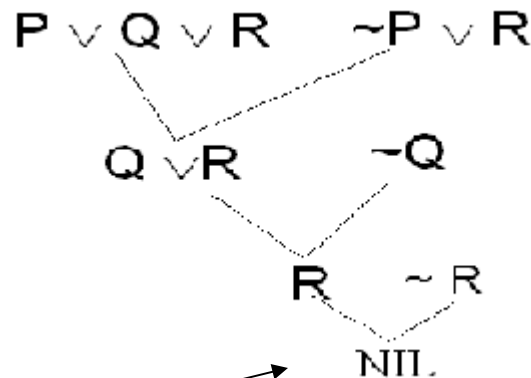
- Method Two (refutation) Graphical representation
 - Insert the negation of the statement as valid sentence

A: $P \vee Q \vee R$

B: $\sim P \vee R$

C: $\sim Q$

D: $\sim R$



An empty clause, which is **false**. This proves the contradiction.

Example: Resolution

- Prove that r follows from:

$$(p \wedge q) \rightarrow (r \vee s) \quad - (1)$$

$$p \rightarrow \sim s \quad - (2)$$

$$p \wedge q \quad - (3)$$

- Solution:

Clause (1) in Clausal form

$$\sim (p \wedge q) \vee (r \vee s)$$

$$\equiv \{ \sim p \vee \sim q \vee r \vee s \} \quad - (1)$$

Clause (2) in Clausal form

$$\{ \sim p \vee \sim s \} \quad - (2)$$

Clause (3) in Clausal form

$$\{ p \} \quad - (3)$$

$$\{ q \} \quad - (4)$$

Assume not r which $\{ \sim r \}$ in Clausal form - (5)

Example: Resolution

Using inference rules: from unit resolution rule of (1) and (5)

$\{\sim p \vee \sim q \vee s\}$ - (6) (resolve r with $\sim r$ and get resolvent)

from unit resolution of (3) and (6)

$\{\sim q \vee s\}$ - (7) (resolve p with $\sim p$ and get resolvent)

from (4) and (7)

$\{s\}$ - (8) (resolve q with $\sim q$ and get resolvent)

from (2) and (8)

$\{\sim p\}$ - (9) (resolve p with $\sim p$ and get resolvent)

from (3) and (9)

$\{\}$ - (10)

Therefore r follows from the original clauses

What rule is used for the conclusion?

1. world population couldn't continues to grow or cities will become hopelessly crowed; Cities will not be hopelessly crowded, or pollution will become intolerable. Therefore, world population couldn't continues to grow or pollution will become intolerable. 7
2. Either Yohanes or Thomas was in Ethiopia; Yohanes was not in Ethiopia. Therefore, Thomas was in Ethiopia.6
3. If twelve million children die yearly form starvation, then something is wrong with food distribution; Twelve million children die yearly form starvation. Therefore, something is wrong with food distribution.1
4. If Japan cares about endangered species, then it has stopped killing whales; Japan has not stopped killing whales. Therefore, Japan does not care about endangered species.1
5. If Napoleon was killed in a plane crash, then Napoleon is dead; Napoleon is dead. Therefore, Napoleon was killed in a plane crash.

Logical equivalence

- Two sentences are **logically equivalent** iff they have the same truth value in all possible world
- Equivalently $\alpha \equiv \beta$ iff $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Logical equivalence

➤ Prove the following using logical equivalence

1. Prove that $(P \vee H) \wedge \neg H \Rightarrow P$ is valid

2. Prove **S**, given that:

$$(P \wedge Q)$$

$$(P \rightarrow R)$$

$$(Q \wedge R) \rightarrow S$$

Converting to CNF

- Converting the following sentence to CNF:

$$a \wedge \sim b \leftrightarrow c$$

$$\equiv (a \wedge \sim b) \leftrightarrow c$$

- Steps:

1. Remove Bi-implication

$$[(a \wedge \sim b) \rightarrow c] \quad \wedge \quad [c \rightarrow (a \wedge \sim b)]$$

2. Remove Implication

$$[\sim(a \wedge \sim b) \vee c] \quad \wedge \quad [\sim c \vee (a \wedge \sim b)]$$

3. Push Negations Inwards

$$[\sim a \vee \sim\sim b \vee c] \wedge [\sim c \vee (a \wedge \sim b)]$$

4. Eliminate Double Negations

$$[\sim a \vee b \vee c] \wedge [\sim c \vee (a \wedge \sim b)]$$

5. Push Disjunctions into Conjunctions

$$[\sim a \vee b \vee c] \wedge [\sim c \vee a] \wedge [\sim c \vee \sim b]$$

Converting to CNF

Convert the following sentence to CNF:

$$((a \rightarrow b) \rightarrow c)$$

- Eliminate Implication

$$\equiv (\sim a \vee b) \rightarrow c$$

$$\equiv \sim(\sim a \vee b) \vee c$$

- Push Negations Inwards

$$\equiv (\sim\sim a \wedge \sim b) \vee c$$

- Eliminate Double Negations, apply De Morgans law

$$\equiv (a \wedge \sim b) \vee c$$

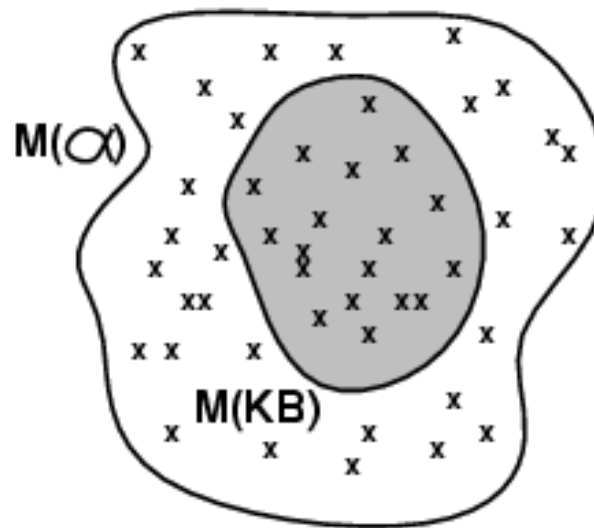
- Push Disjunctions into Conjunctions

$$\equiv (a \vee c) \wedge (\sim b \vee c)$$

- Hence $(a \vee c) \wedge (\sim b \vee c)$ is CNF of $((a \rightarrow b) \rightarrow c)$

Model

- Any world in which a sentence is true under a particular interpretation is called model of that sentence under that interpretation
- We say m is a model of a sentence α if α is true in m
- $M(\alpha)$ is the set of all models of α
- Then $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$
 - ▶ E.g. $KB = \text{Giants won and Reds won}$ $\alpha = \text{Giants won}$



Complexity of propositional inference

- The worst case is exponential (2^n).
- In this case we can prove using truth table
- Using inference rule we may minimize the time complexity.
- The use of inference rules depends on general principles of logic called mono-tonicity
- The property of mono-tonicity of a logic is

if $KB_1 \models \alpha$ then $(KB_1 \cup KB_2) \models \alpha$